Instrumentation Amplifier: Phase Two

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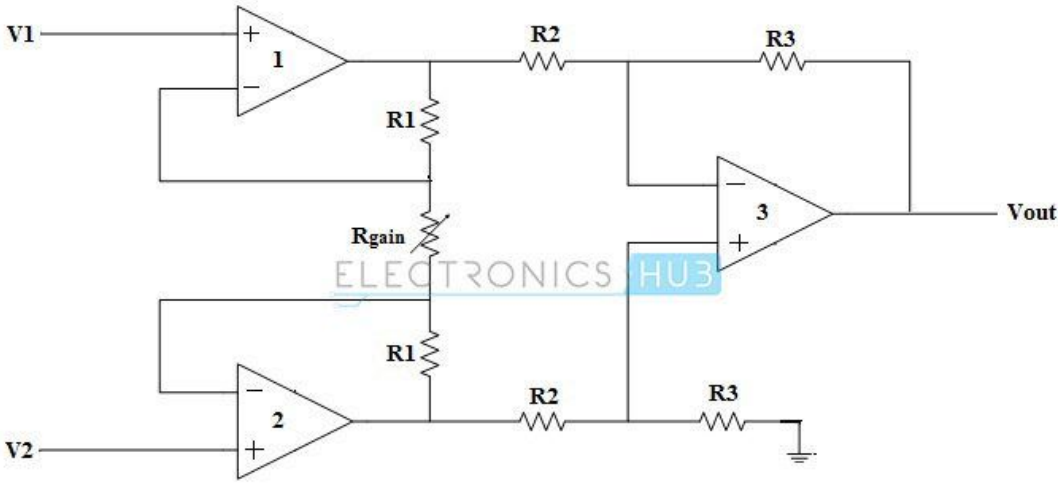
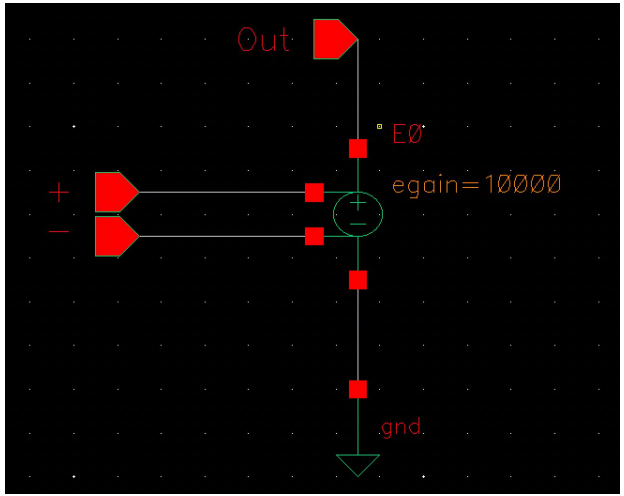
Topology

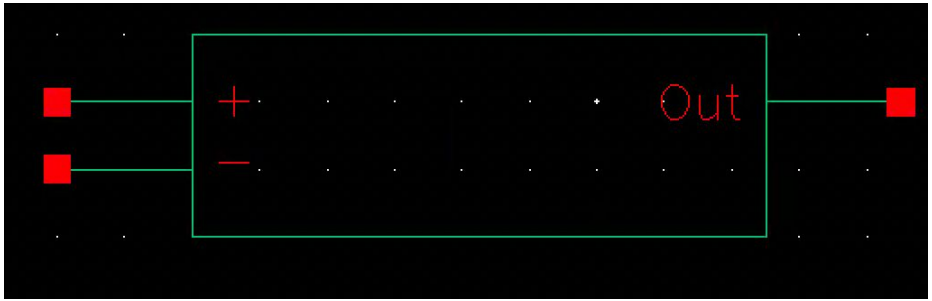
- $V_{out} = (R_3/R_2)[(2R_1 + R_{gain})/R_{gain}](V_1-V_2)$
- If all of the resistors besides R_{gain} are the same value R: $V_{out} = (V_1-V_2)(1+2R/R_{gain})$



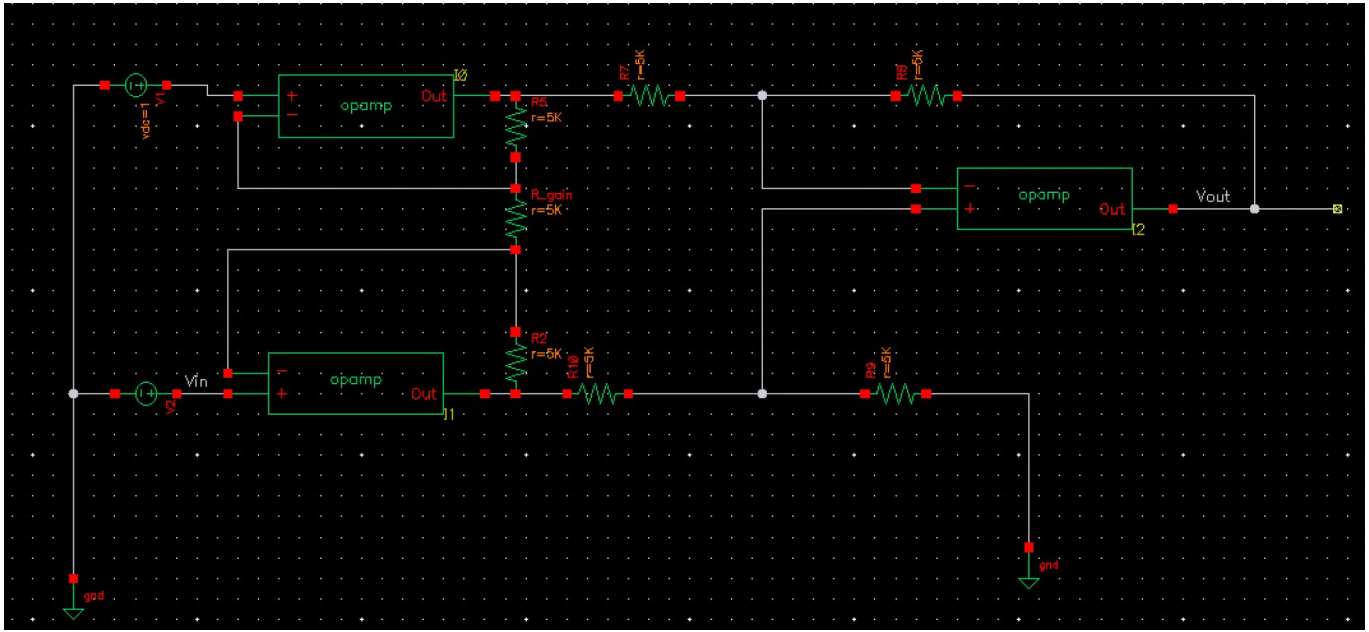
A Typical Three Op-amp Instrumentation Amplifier

Op-Amp Schematic and Symbol



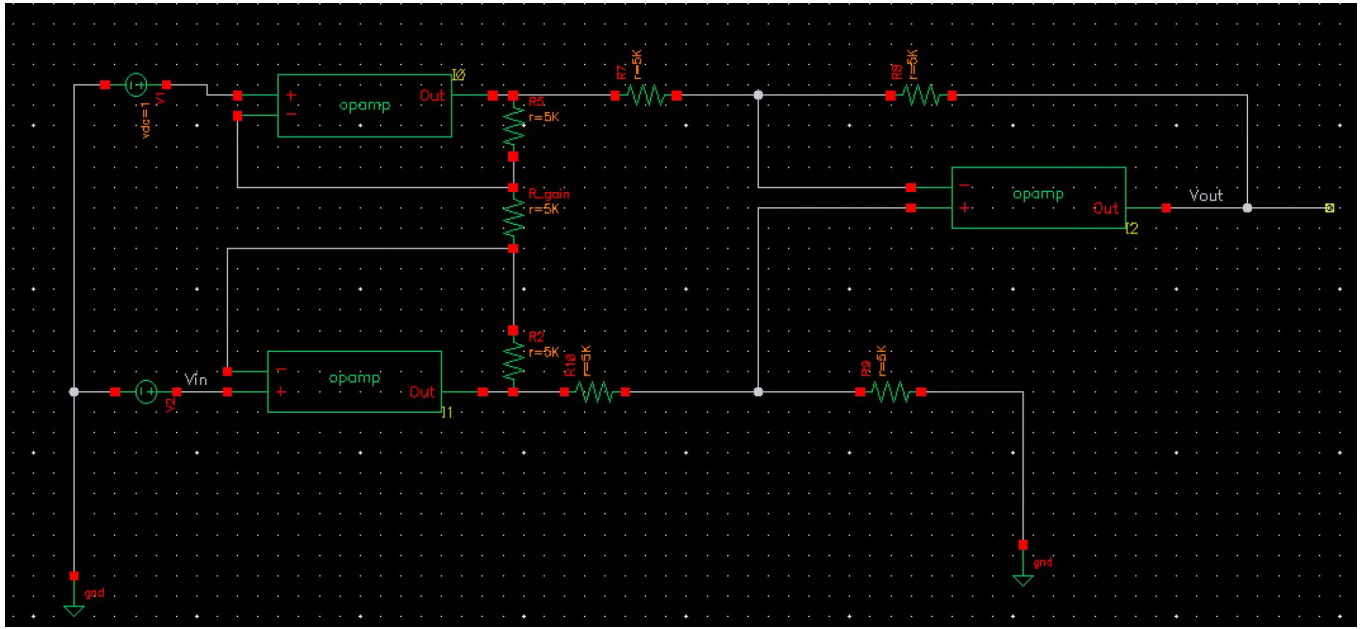


Instrumentation Amplifier Schematic



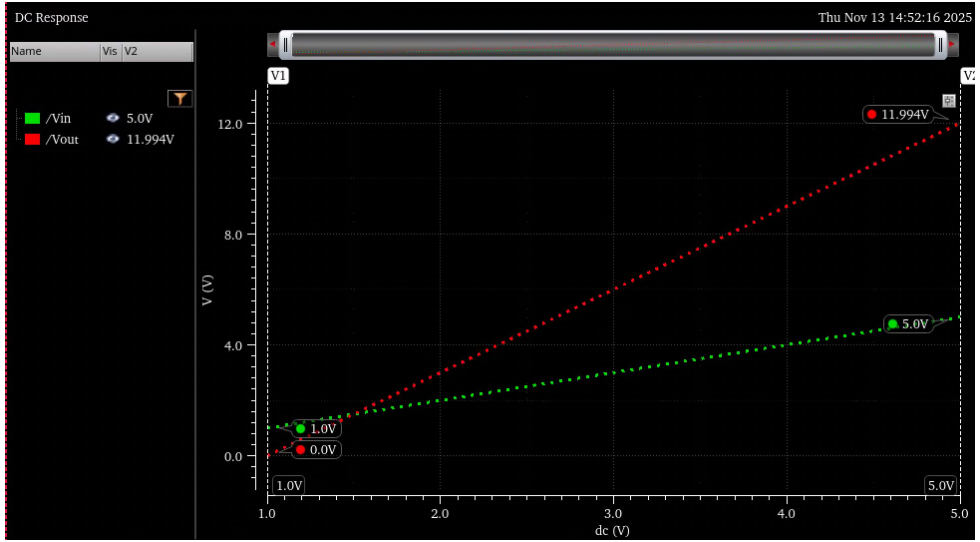
Testing differential-mode gain

1 input held to 1 V while the other input is swept from 1 V to 5 V.



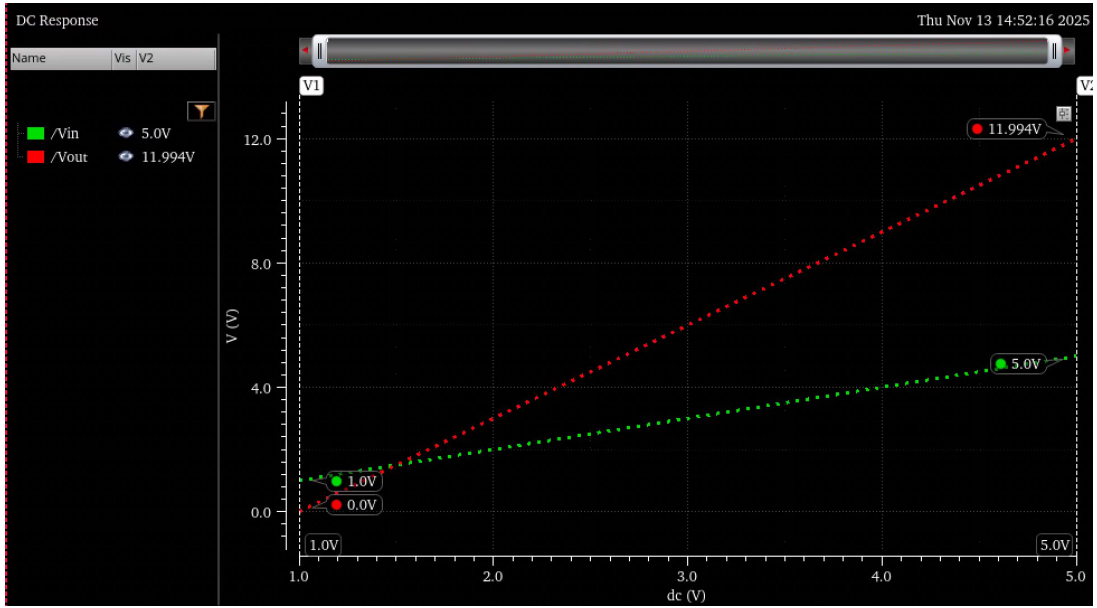
Testing differential-mode gain

- 1 input held to 1 V while the other input is swept from 1 V to 5 V and R_gain = R
- $V_{out} = (V_1 - V_2)(1+2R/R_{gain}) = (V_1 - V_2)(3)$



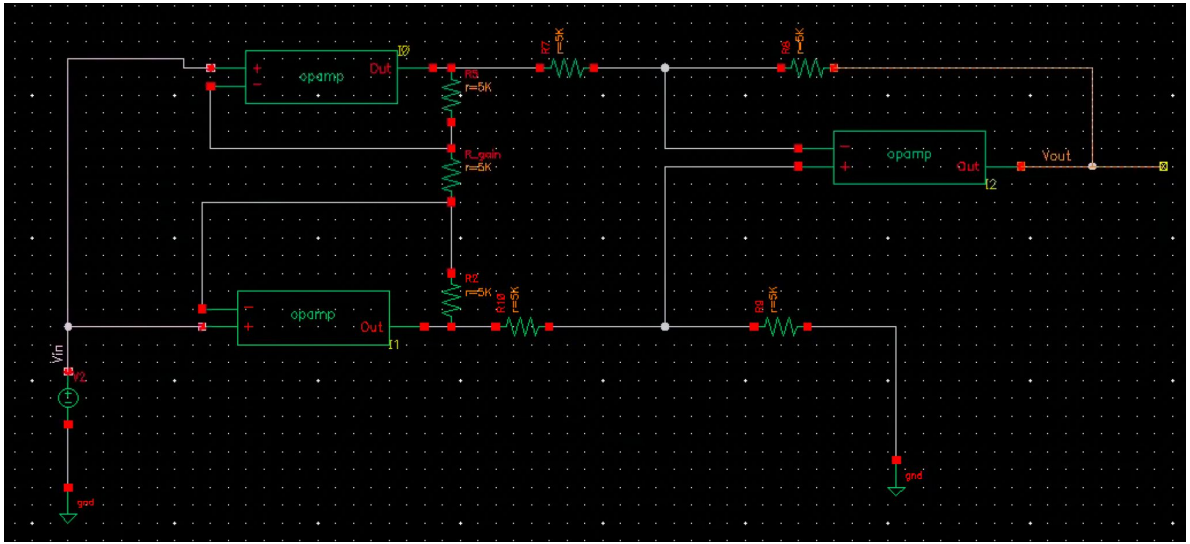
Testing differential-mode gain

- $V_{out} = (V_1 - V_2)(1+2R/R_{gain})$
  $= (V_1 - V_2)(3)$
- $A_{DM} = 3$
- Simulation results are equal to calculated results.
  - When $V_1 = V_2$: $V_{out} = 0$
  - When $V_1 - V_2 = 4$, $V_{Out} = 12$



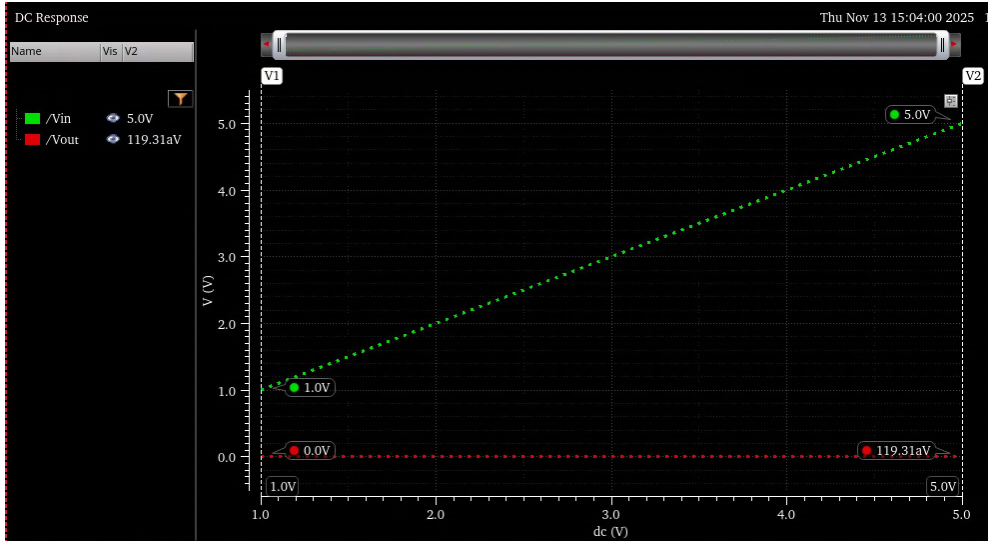
Testing common-mode gain

- Providing the same voltage to both inputs
- Expectation: $V_{out} = 0$ because $V_1 - V_2 = 0$
  - $A_{CM} = 0$



Testing common-mode gain

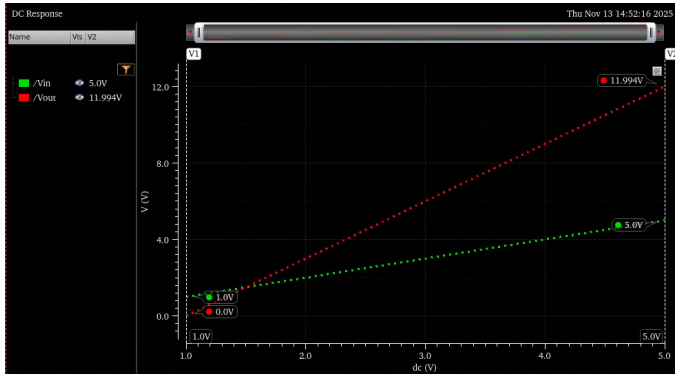
- Expectation: $V_{out} = 0$ because $V_1 - V_2 = 0$
- Results: $V_{out} = 0$



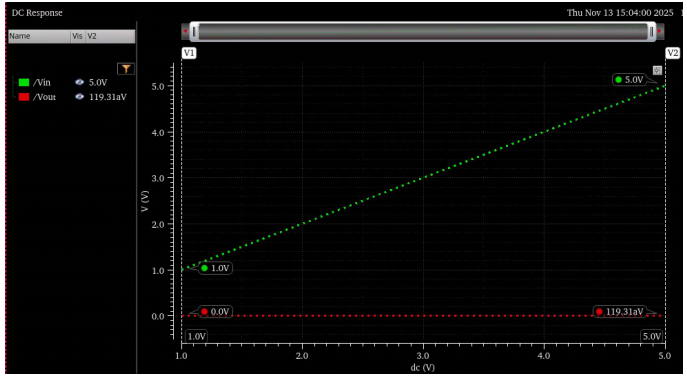
CMRR

- Combining both the differential mode and common-mode examples:
  - CMRR = $20\log(A_{DM}/A_{CM}) = 20\log(3/0) = \infty$ (ideal)

Differential mode

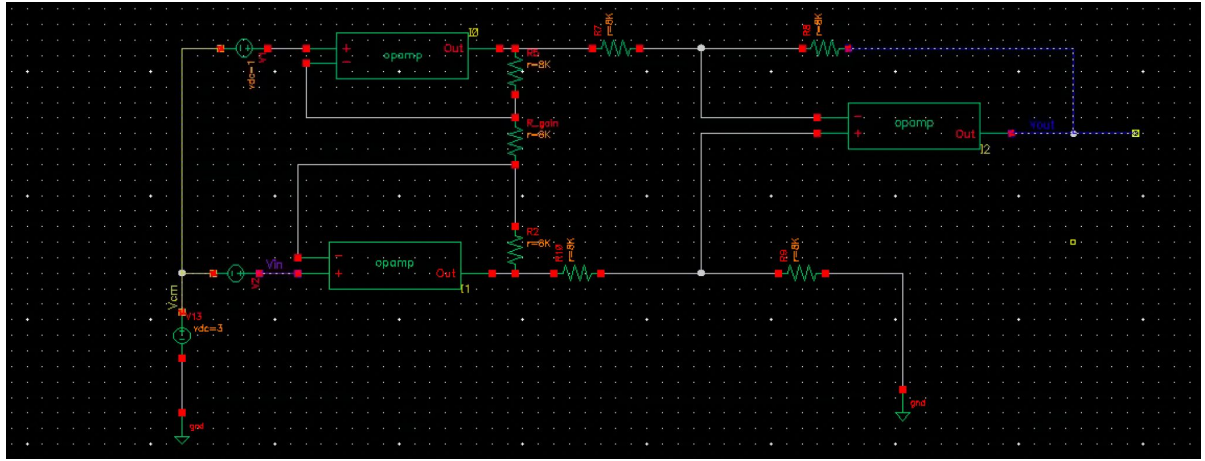


Common mode



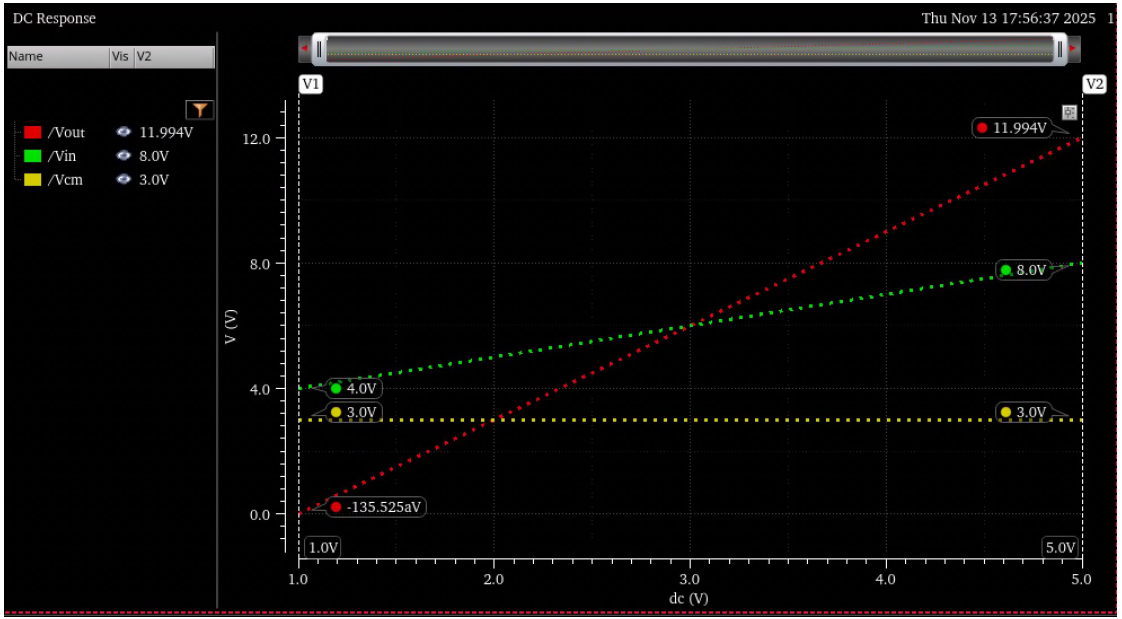
Another calculation of CMRR (with a common-mode and differential-mode input)

- $R = R_{gain} = 8\ k\Omega$: $V_{out} = (V_1 - V_2)(1+2R/R_{gain}) = (V_1 - V_2)(3)$
- For a differential amplifier in general:
  - $V_{out} = A_{dm} vi_{i,\ dm} + A_{CM} V_{I,\ CM}$
- Simulation with $V_{CM} = 3$ V and sweeping $V_1 - V_2$ from 0 to 4 V



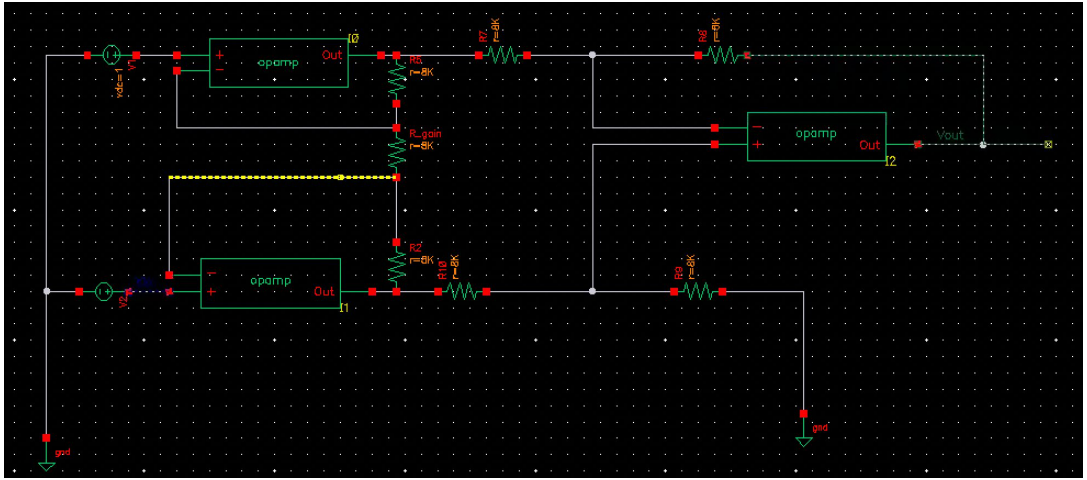
Another calculation of CMRR (with a common-mode and differential-mode input) (continued)

- For a differential amplifier in general: $V_{out} = A_{dm} vi_{i,\ dm} + A_{CM} V_{I,\ CM}$
- Simulation with V_{CM} = 3 V and sweeping V_1 - V_2 from 0 to 4 V
- From the graph: V_{out} is still equal to $(V_1 - V_2)(1+2R/R_{gain}) = (V_1 - V_2)(3)$



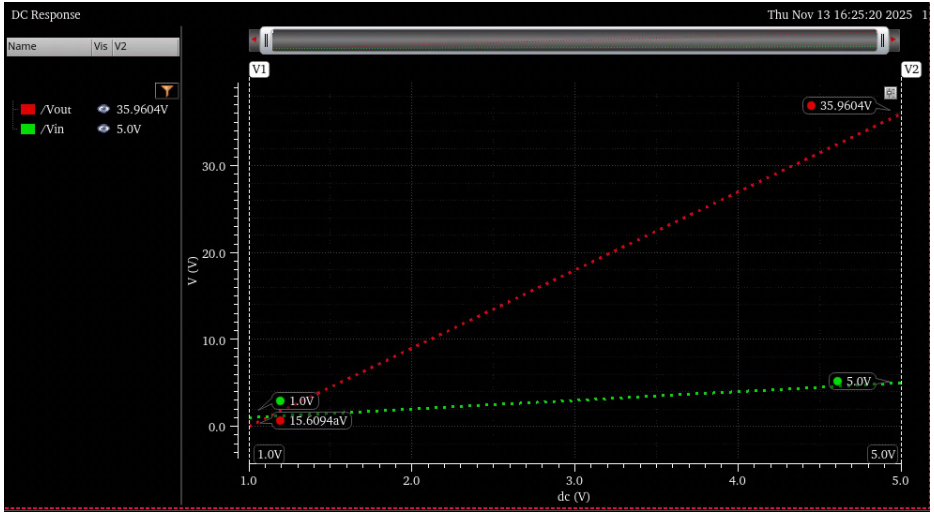
Adjusting R_{gain}

- $V_{out} = (V_1-V_2)(1+2R/R_{gain})$
  - V_{out} can be adjusted by varying R_{gain}
- One input is 1 V and the other input is swept from 1 V to 5 V
- Using R = 8k and varying R_{gain}:
- Expected values
  - R_{gain} = 2 kΩ: $V_{out} = (V_1-V_2)(9)$
  - R_{gain} = 4kΩ: $V_{out} = (V_1-V_2)(5)$
  - R_{gain} = 8kΩ: $V_{out} = (V_1 - V_2)(3)$



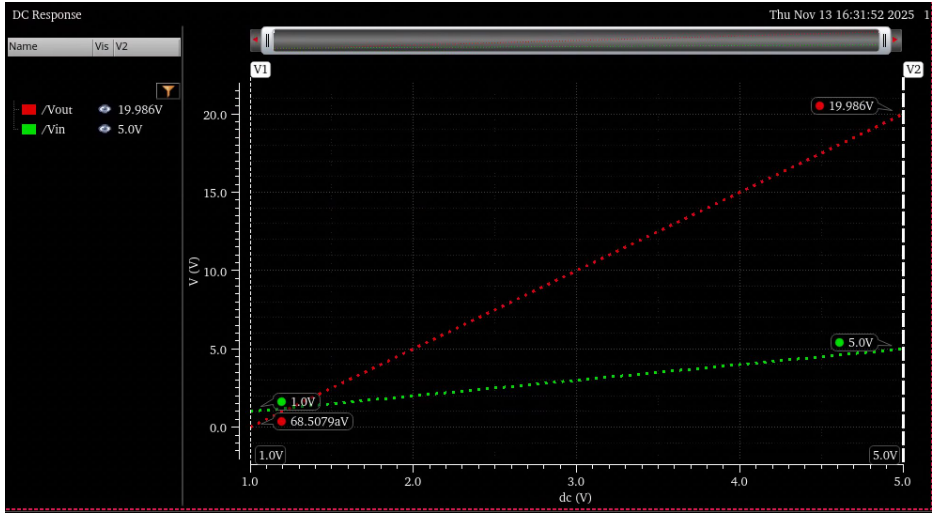
Adjusting R_{gain}: R_{gain} = 2kΩ

- One input is 1 V and the other input is swept from 1 V to 5 V
- R_{gain} = 2 kΩ: $V_{out} = (V_1 - V_2)(9)$
- Expected value is the same as the simulated value:
  - At 1 V: $V_{out} = (0)(9) = 0$ V
  - At 5 V: $V_{out} = (4)(9) = 36$ V



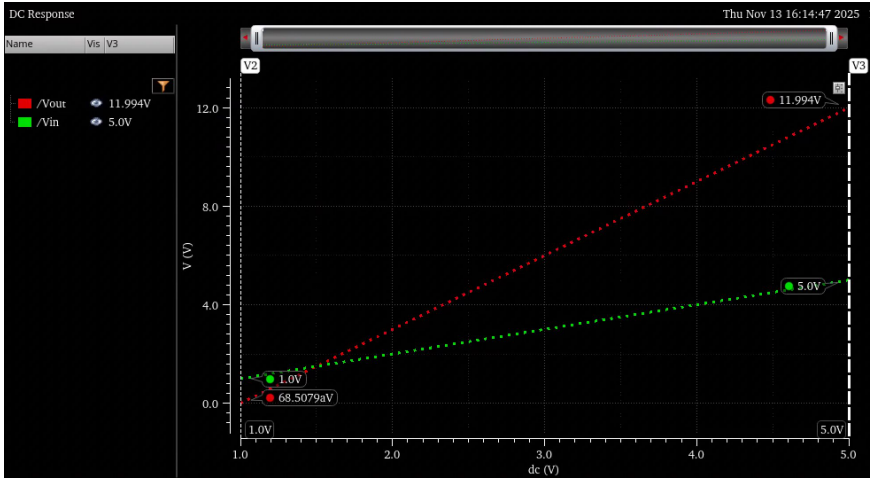
Adjusting R_{gain}: R_{gain} = 4kΩ

- One input is 1 V and the other input is swept from 1 V to 5 V
- R_{gain} = 4 kΩ: $V_{out} = (V_1 - V_2)(5)$
- Expected value is the same as the simulated value
  - At 1 V: $V_{out} = (0)(5) = 0$ V
  - At 5 V: $V_{out} = (4)(5) = 20$ V



Adjusting R_{gain}: R_{gain} = 8kΩ

- One input is 1 V and the other input is swept from 1 V to 5 V
- Expected: R_{gain} = 8kΩ: $V_{out} = (V_1 - V_2)(3)$
- Expected value is the same as the simulated value
  - At 1 V: $V_{out} = (0)(3) = 0$ V
  - At 5 V: $V_{out} = (4)(3) = 12$ V



Sources

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- https://www.youtube.com/watch?v=INeLsIoHw7A